Do Now · retrieval

Lesson 048 · C# List<T>: Add, Insert and Remove

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does Array.Length give?

Q2

What is zero-based indexing?

Q3

What is aliasing between references?

Q4

A rising-edge D-type flip-flop starts with Q=0. D rises to 1 and then returns to 0 entirely between two rising clock edges. Explain why Q remains 0.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — List<T> is a resizable sequence. Add appends; Insert inserts at a specified index and shifts following items.

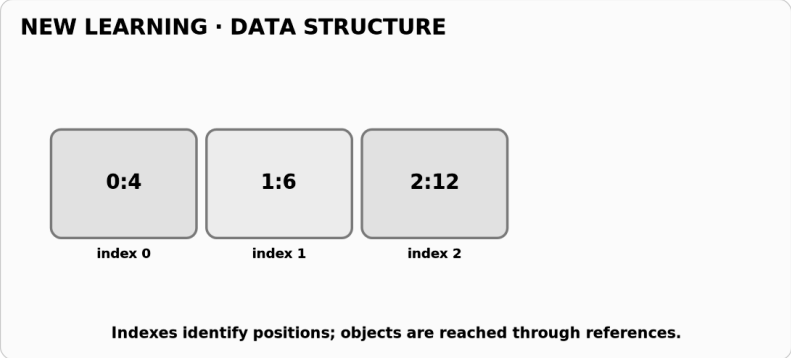
Core idea 2

Explain — RemoveAt removes an item by index; Remove(value) removes the first matching value instead.

Core idea 3

Apply — Unlike an array, List<T> changes Count as items are inserted/removed. An index must still be within the valid bounds.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

List<T> — Resizable generic sequence.

Insert — Adds at a specified index.

RemoveAt — Removes item by index.

Count — Number of items currently stored.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

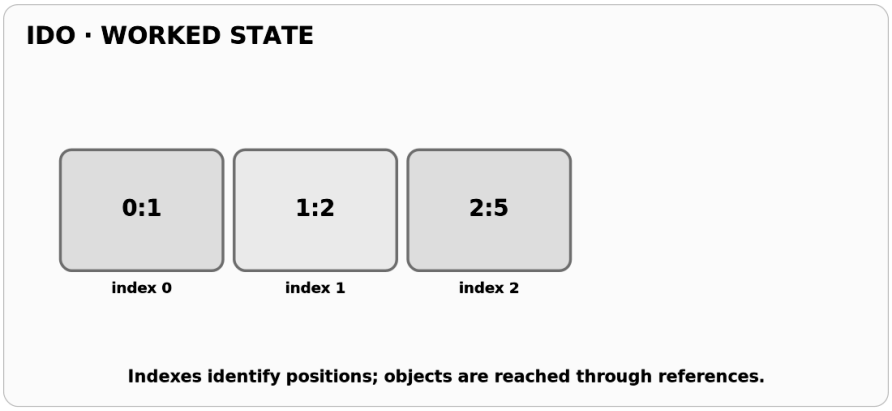
Problem

Trace a list after inserting at the start, then removing the last element.

Follow the worked method

1. Trace and justify — After Insert: [1,2,5,7]; RemoveAt(3) removes 7; final [1,2,5].

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Trace and justify
[3,9,12]. Add appends 12; Remove(6) removes matching value 6, unlike RemoveAt(6).

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace Insert(1,6) then RemoveAt(2) for [4,8,12]. [6]

Question 2

Explain the difference between Remove(5) and RemoveAt(1) for [2,5,5,9]. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

For [1,2,3], write C# that inserts 0 at start, appends 4 and outputs Count; state why List<T> suits a growing collection. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

RemoveAt(5) means remove the value 5.

Use key vocabulary from memory

1. Explain List<T> in your own words.

2. Explain Insert in your own words.

3. Explain RemoveAt in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.